

Iteration I

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High level Project Analysis

- What is the System about?
 - Booking/Management software
- Actors: Guest, Clerk, Admin
- What is the Problem?
 - Guests want to book rooms
 - Admins want to manage rooms

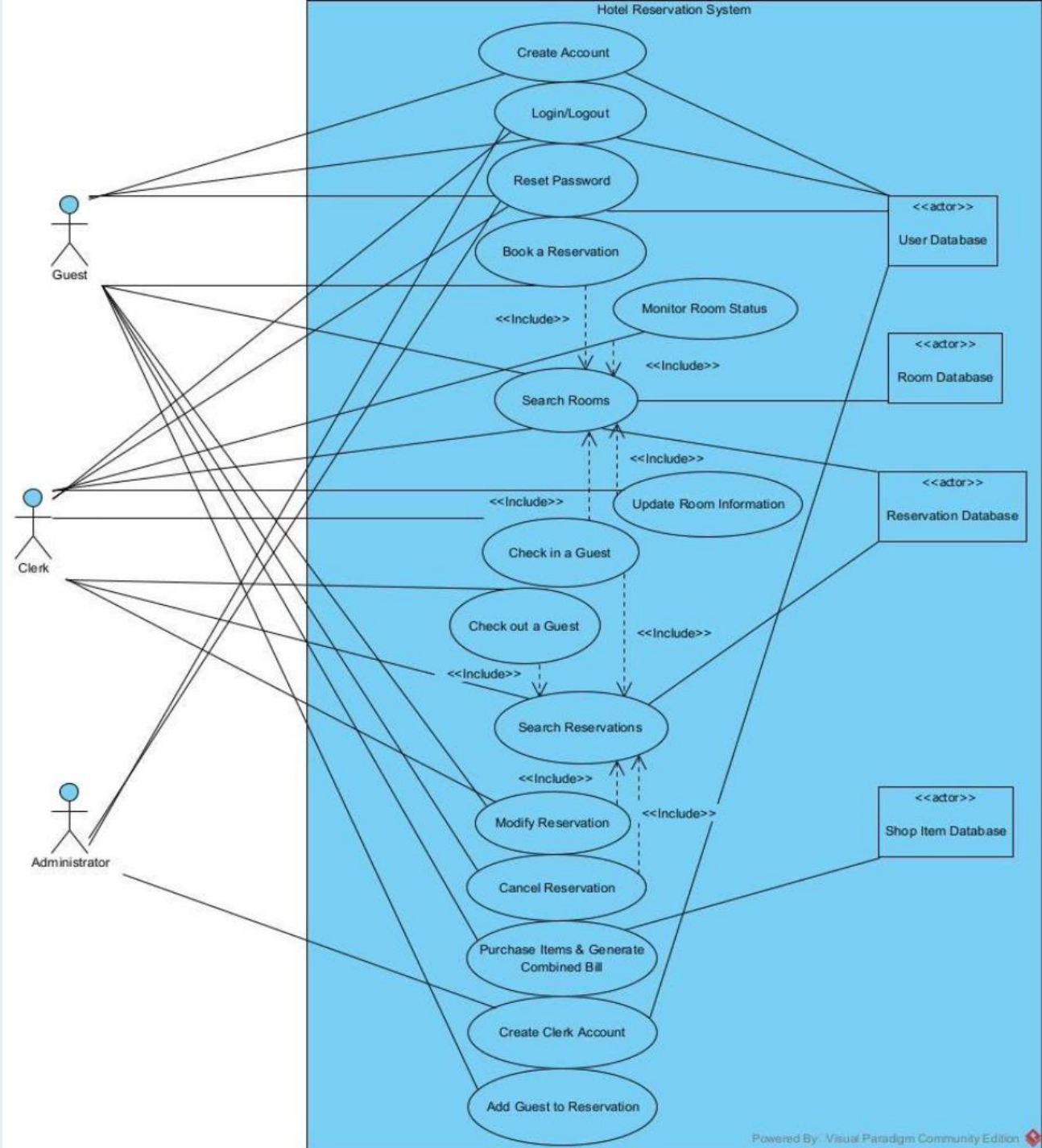


Requirements and Use Cases



- Key Functional Requirements:
 - Functionalities for Guests
 - Ex: Book room, modify reservation, cancel reservation, etc.
 - Functionalities for Clerk
 - Check in guests, update room information, etc.
 - Functionalities for Admin
 - Manage rooms, manage system information, etc.

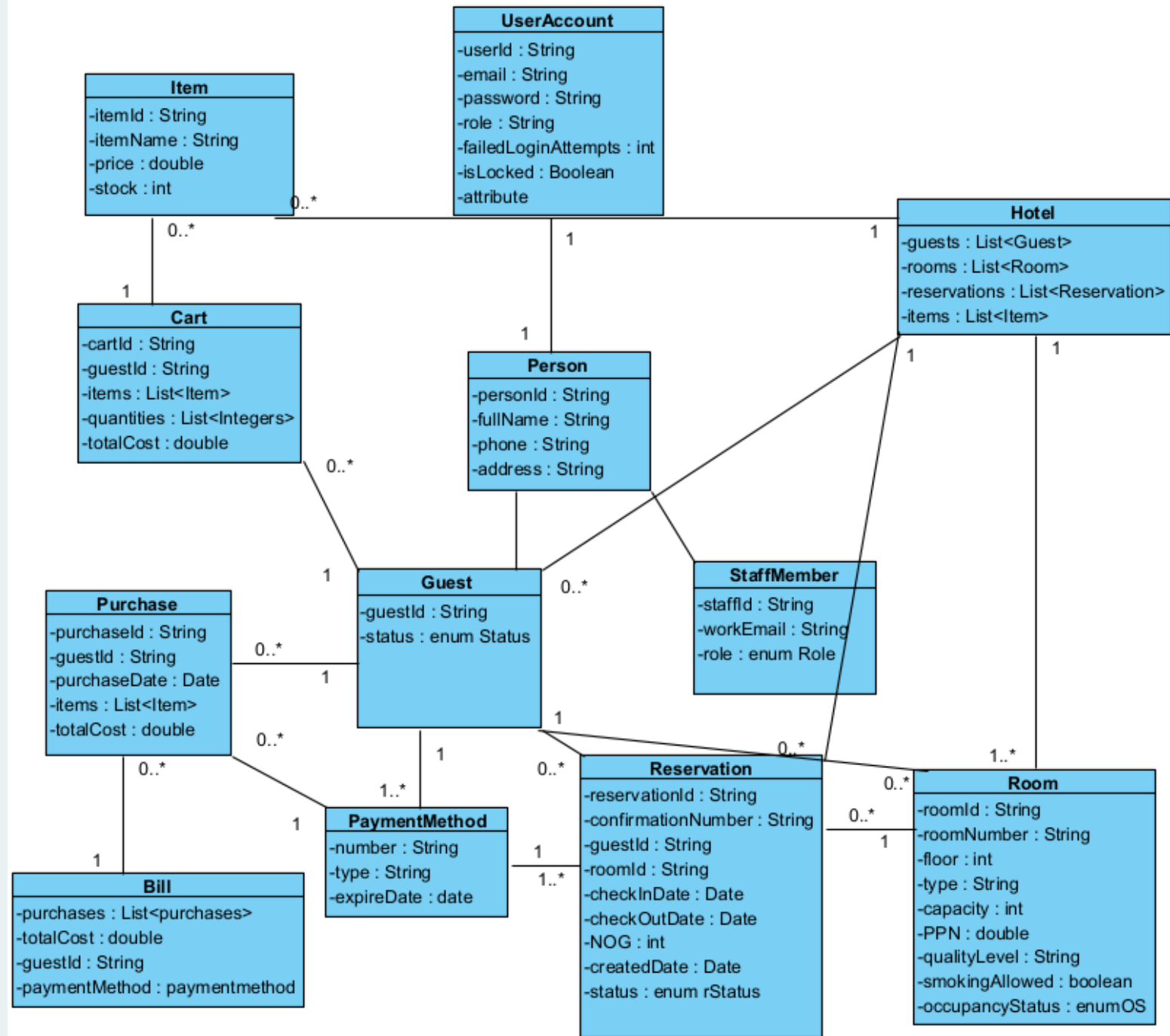
Use Case Diagram



Domain Model + SSD + Operation Contracts



Domain Model



Use Cases: Book a Reservation

Use Case Name	Book a Reservation
Use Case Author	Hannah Ross
Actor	Hotel Guest
Preconditions	1. <u>Guest is</u> logged in. 2. The system shows available rooms.
Postconditions	1. A reservation is saved. 2. Room is reserved for specified dates.
Main Success Scenario	1. Choose Guest selects desired room. 2. System requests Guest Info (Name, Address, Credit Card, Stay Dates). 3. <u>Guest enters</u> required details. 4. System validates credit <u>card</u>, <u>address</u>, and availability. 5. System calculates stay cost and provides summary of information. 6. Guest approves <u>cost amount</u> and details of <u>reservation</u>. 7. System saves <u>reservation</u>. 8. System displays <u>confirmation number</u>.
Extensions	1. Invalid search criteria (step 4) i. The system detects invalid input (e.g., check-out date earlier than check-in date). ii. The system displays an error message and requests corrected input.

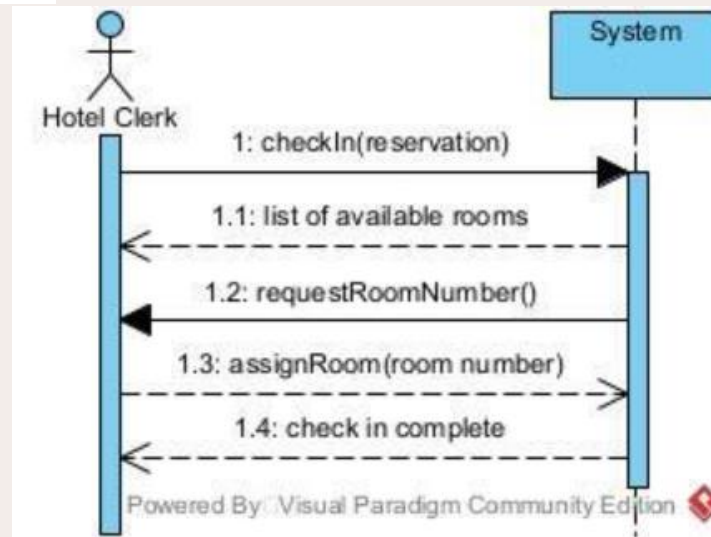
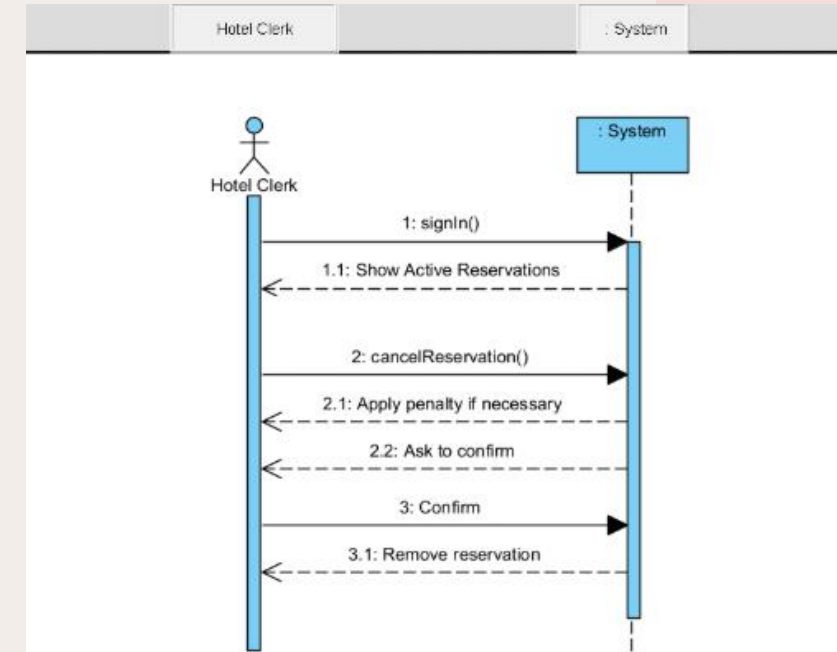
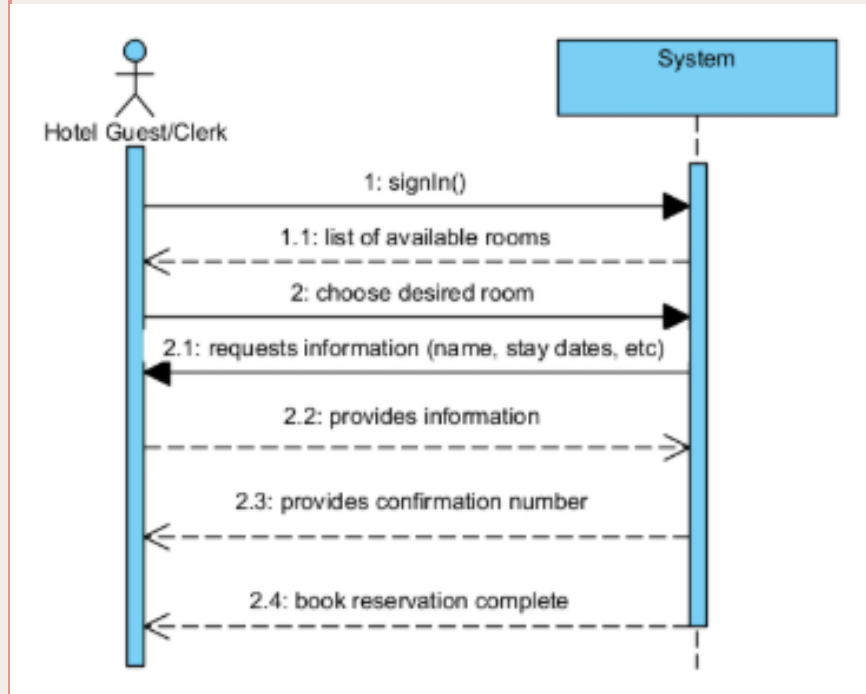
Use Cases: Check in a Guest

Use Case Name	Check in a Guest
Use Case Author	Matt Freeman
Actor	Hotel Clerk
Preconditions	1. The clerk is logged in.
Postconditions	1. The guest's status is set to " <u>checked-in</u> ." 2. A room number is assigned to the guest and that room is set to "occupied."
Main Success Scenario	1. A guest approaches the clerk to check in. 2. The clerk searches for the guest's reservation by name or confirmation number. 3. The system <u>displays the reservation</u> and room specifications. 4. The clerk selects "Check-In." 5. The system displays available rooms matching the <u>reservation's</u> room specifications. 6. The clerk selects "Assign Room" <u>on a chosen</u> room. 7. The system sets the guest to "checked-in" and the room number to "occupied."
Extensions	1. No Reservation Today (step 2) i. The guest does not have a reservation today ii. The system displays an error message 2. No Clean Rooms (step 5) i. No clean rooms matching the specifications can be displayed. ii. The clerk may override this and search for other rooms to assign.

Use Cases: Cancel Reservation

Use Case Name	Cancel Reservation
Use Case Author	Duane Schlottke
Actor	Hotel Guest / Clerk
Preconditions	1. Valid reservation exists prior to start date. 2. User is logged in 3. The cancellation occurs 2+ days to <u>reservation</u> date.
Postconditions	1. System deletes reservation on Guest's account 2. System updates room to available
Main Success Scenario	1. User selects a reservation to cancel. 2. System checks <u>date</u> of original reservation creation. 3. If > 2 days since creation, system calculates 80% single-night penalty. 4. System displays total refund/penalty and requests confirmation. 5. User confirms. 6. System updates status to Cancelled
Extensions	1. User Aborts (step 5) i. Cancellation is canceled; reservation remains active.

System Sequence Diagrams



Operation Contracts

Contract CO1: bookReservation

Operation: bookReservation(reservation)

Cross References: Use Case – Sign in, Check in a Guest

Preconditions

1. The system is online
2. Guest is authenticated
3. At least one room = “Available” for requested dates/other details.

Postconditions

1. The room is selected and moved to “reserved/booked/occupied”
2. The reservation is aligned with the correct guest and room.
3. The confirmation number is generated.
4. The system updates the reservation with details.

Contract CO1: cancelReservation

Operation: cancelReservation(room)

Cross References: Use Case – Sign in (clerk, guest)

Preconditions

10. The system is online.
11. Clerk/Guests authenticated.
12. Room to be selected is valid reservation

Postconditions

9. The room is found in the system.
The system cancels reservation

Contract CO1: checkIn

Operation: checkIn(reservation)

Cross References: Use Case – Check in a Guest

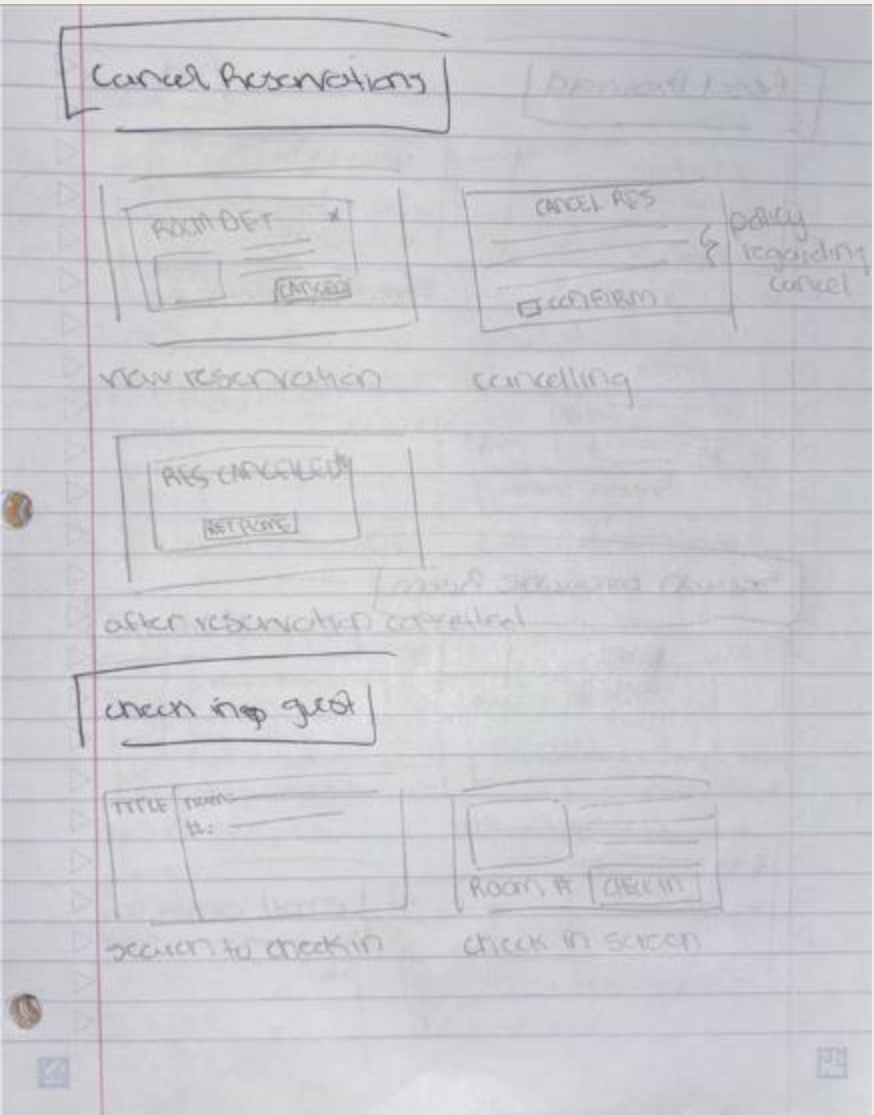
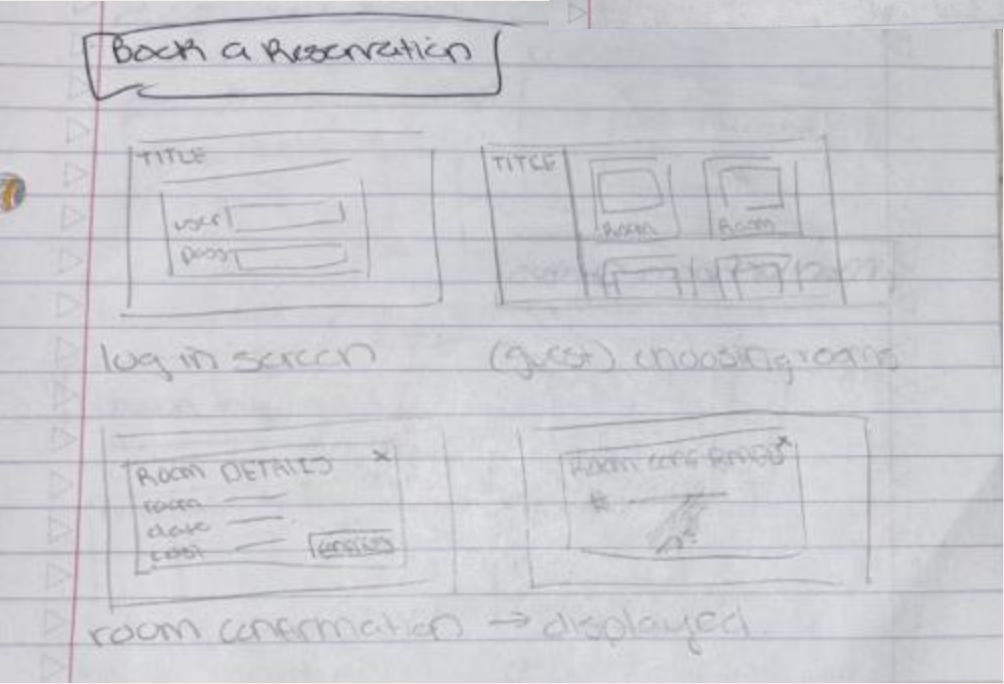
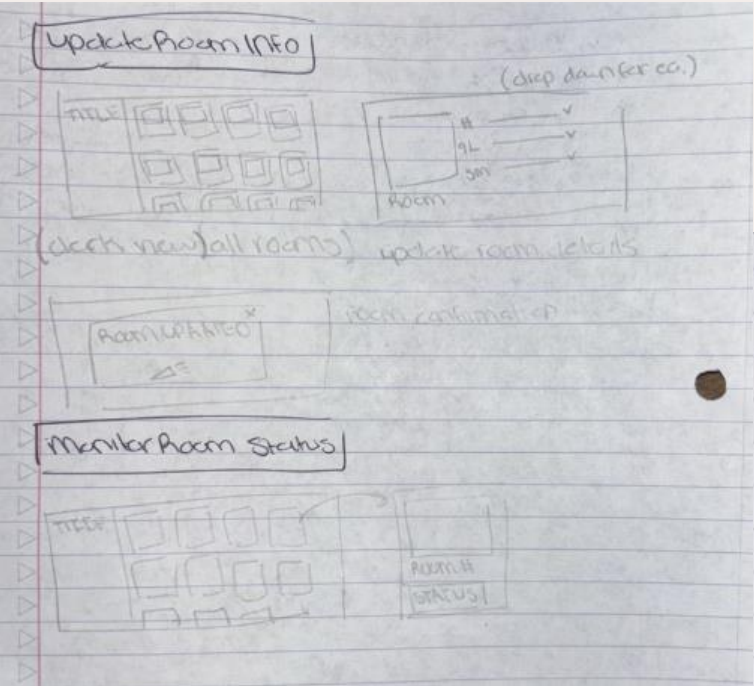
Preconditions

20. The system is online
21. Room and reservation data exist

Postconditions

11. The guest’s status is updated to “checked-in” and assigned a room number
12. The reservation’s status is updated to “checked-in” and assigned a room number
13. The room is set to “occupied”

UI Wireframes



Project Management Metrics



Name	Role	Hours
Matt Freeman	Librarian	8
Emmanuel Humber	Quality Assurance Engineer	7
Andrew Hutcheson	Requirements Engineer	7
Hannah Ross	Design Engineer	7
Duane Schlottke	Project Manager	7

Backlog

...

Design Visual themes for each of the floors

determine criteria of quality levels

system of inputting information (which room , days, name for reserving, etc)

discuss corner cases (trying to book rooms already taken, system response when hotel is full, etc)

+ Add a card

User Stories

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Account Creation: As a New Guest, I want to create an account with a username and password, so that I can manage my personal details and future bookings.

Room Search: As a Guest, I want to search for available rooms by date and theme (Nature, Urban, Vintage), so that I can find a stay that matches my preference.

Booking: As a Guest, I want to make a reservation by providing my stay dates, room type, and payment info, so that my room is guaranteed for my visit.

Modification: As a Guest, I want to modify my existing reservation, so that I can adjust my travel plans without starting over.

Cancellation: As a Guest, I want to cancel my reservation, so that I can release the room if my plans change.

Product Browsing: As a Guest, I want to browse local artisanal goods and clothing.

Sprint

...

+ Add a card

In Progress

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Andrew - Make Website

distinctions between admin and clerk

Fix inconsistencies across use cases

+ Add a card

Completed

...

✔ Define Project Vision

✔ Matt - Make Use Case Diagram

✔ Gantt Design

✔ Hannah - Make UI Wireframes

✔ Duane - Make Powerpoint

✔ Matt - Make Iteration 1 PDF Deliverable

✔ Everyone - Finish 3 Fully Dressed Use Cases, Operation Contracts, and SSDs

✔ Gantt Chart Design

✔ Project Anylsis

✔ Emmanuel - Make Domain Model

+ Add a card

Thank
you

